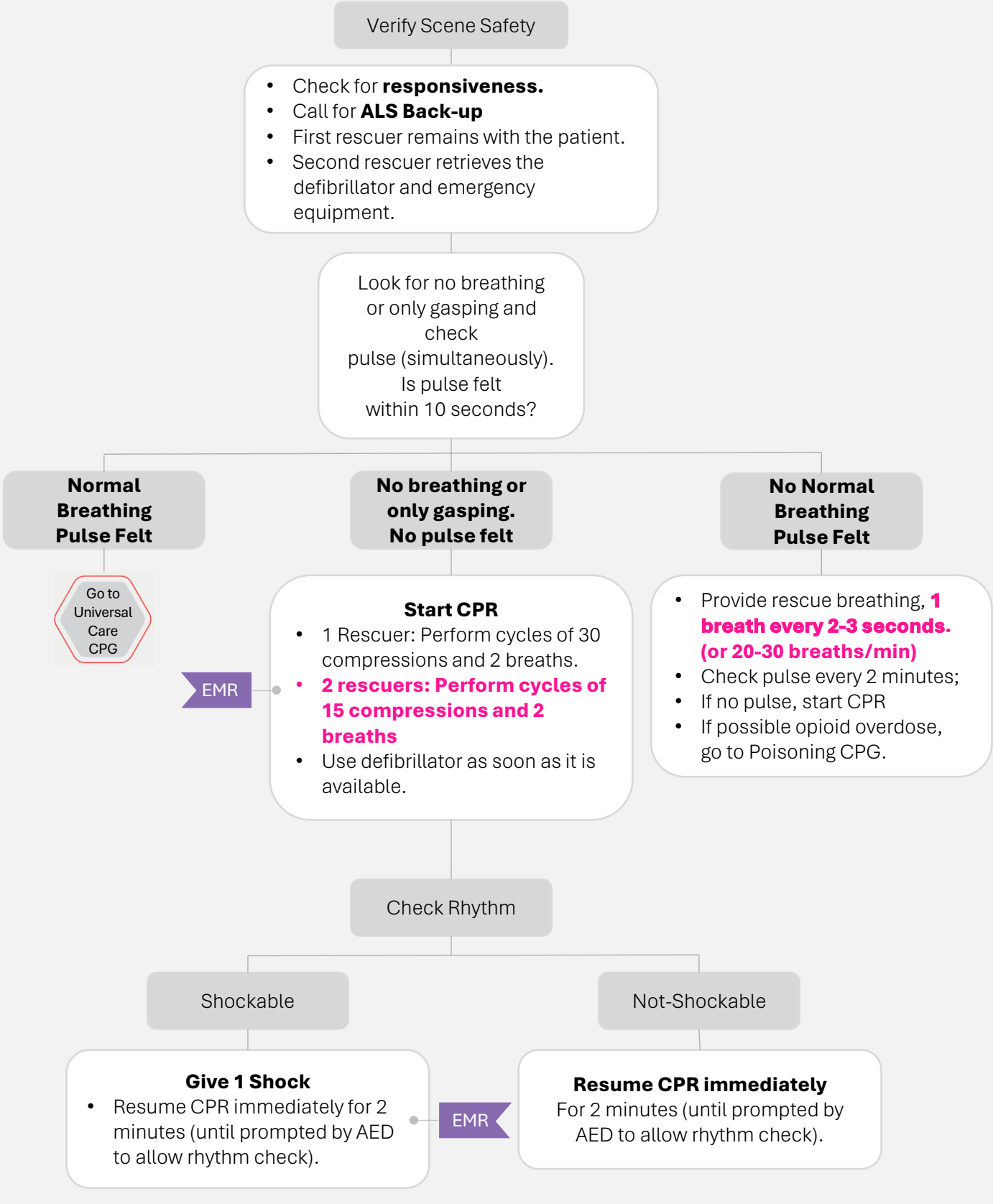
Pediatric Basic Life Support (<13 years)



Use Adult pads if patient is:
- >13 years old, or
- >55kg

Patient Care Goals

1. Recognize cardiac arrest promptly in infants and children: Quickly identify unresponsiveness, abnormal or absent breathing, and absence of a pulse or inadequate perfusion to initiate BLS without delay.
2. Provide immediate, high-quality chest compressions and effective ventilations: Deliver compressions at the correct depth and rate, with full chest recoil, and minimize interruptions; combine with effective ventilations using age-appropriate techniques.
3. Minimize interruptions in compressions and optimize oxygen delivery: Keep interruptions in compressions under 10 seconds, prioritize maintaining adequate oxygenation through timely ventilations, and prevent hypoxia.
4. Use defibrillation promptly if indicated: Recognize shockable rhythms early, attach an AED as soon as available, and deliver a shock without delay when indicated.
5. Maintain normoxia and avoid hyperventilation: Provide just enough oxygen to achieve an SpO_2 ≥94%, avoiding unnecessary hyperoxia; ventilate at an appropriate rate and volume to prevent hyperventilation and maintain $EtCO_2$ in the normal range (35–45 mmHg).

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Infants and children ≤13 years found unresponsive, not breathing normally, and/or pulseless.
- Infants and children with inadequate circulation require BLS interventions.

Exclusion Criteria

- Obvious signs of irreversible death (e.g., rigor mortis, dependent lividity, decapitation).

Key Patient Assessment Points

1. **Scene Safety & Initial Impression**
   - Confirm scene is safe. Assess for responsiveness using tactile and verbal stimuli.
2. **Breathing Assessment**
   - Check for normal breathing (no more than 10 seconds). Gasping is not normal breathing.
3. **Circulation**
   - Check central pulse (carotid in children >1 year, brachial in infants <1 year) within 10 seconds.
   - If no pulse or HR <60 bpm with poor perfusion despite oxygenation and ventilation, begin chest compressions.
2. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - All unconscious patients and patients in cardiac arrest are Red Priority.

Patient Management & Interventions

1. **Check for Responsiveness**
   - Verify unresponsiveness by assessing for no response to verbal or painful stimuli (AVPU).
2. **Request ALS back-up**
   - Request ALS support within 5 minutes of arrival if not already activated.
   - Specify "pediatric cardiac arrest" and confirm ETA of ALS unit.
3. **Retrieve Emergency Equipment**
   - First rescuer remains with the patient.
   - Second rescuer retrieves the AED and other emergency equipment.
3. **Primary Survey**
   - Open airway using head-tilt-chin-lift (if no trauma) or jaw-thrust (if trauma suspected).
   - Look for no breathing or only gasping and check pulse (simultaneously).
     - Gasping is not normal breathing.
     - Check central pulse (carotid in children >1 year, brachial in infants <1 year)
6. **No Normal Breathing. Pulse Felt**
   - **Provide Rescue Breathing**
     - 1 breath every 2-3 seconds. (or 20-30 breaths/min)
   - **Check pulse every 2 minutes;**

        - If no pulse, or HR <60 bpm with poor perfusion despite oxygenation and ventilation, start CPR.
    - If possible opioid overdose, go to Overdose/Poisoning CPG.
4. **No Breathing or Only Gasping. No Pulse Felt**
    - **Start CPR**
        - **Perform Chest Compressions**
            - Depth: At least 1/3 the anterior-posterior chest diameter (~4 cm in infants, ~5 cm in children).
            - Rate: 100–120 per minute.
            - Allow full chest recoil between compressions.
            - Minimize interruptions (<10 seconds).
        - **Compression : Ventilation Ratio**
            - If single rescuer: 30:2
            - If two or more rescuers: 15:2
        - **Airway & Ventilation**
            - Open airway and provide 2 effective breaths after every set of compressions.
            - Use BVM with 100% oxygen as soon as available.
        - **Apply AED**
            - Apply AED as soon as it is available.
            - Follow AED instructions.
            - If no Pediatric Pads are available, use Adult Pads.
        - **Deliver Shock if Advised**
            - If shock is advised:
                - Give 1 Shock
                - Resume CPR immediately for 2 minutes (until prompted by AED to allow rhythm check).
        - **Blood glucose**
                - Check blood glucose during resuscitation. If hypoglycemia is identified, treat as per Hypoglycemia CPG.
        - **Continue CPR until ALS Back-up arrives**
        - **Check for signs of ROSC**
            - Check for signs of ROSC (e.g., spontaneous pulse, normal breathing) every 2 minutes.

Patient Safety Considerations

- Avoid excessive ventilation — target normocapnia.
- Ensure correct compression depth & rate.
- Minimize interruptions in CPR.
- Use appropriate-sized equipment for infants/children.
- Defibrillate as soon as possible if shockable rhythm detected.
- Protect airway and cervical spine if trauma suspected.

Documentation

- Identification: Patient's name (if known), age, sex, estimated weight.
- Medical complaint: Cardiac arrest — specify if witnessed, unwitnessed, or found unresponsive.
- Information/Injuries: Relevant history (e.g., trauma, drowning, pre-existing conditions, suspected cause).
- Signs: Initial and ongoing HR, rhythm, SpO_2, $EtCO_2$, skin color, responsiveness.
- Treatment: CPR (start time, quality, depth, interruptions), defibrillation (number, energy), airway interventions, oxygen, glucose, medications.
- Allergies: Any known allergies if identified.
- Medications: Any medications administered during resuscitation.
- Background & Other: Medical history if available, time of collapse, bystander CPR, name & designation of receiving hospital staff.

Key Performance Indicators

1. Initiate CPR within 1 minute of recognition of cardiac arrest.

Clinical Insights & Additional Guidance

- Pediatric cardiac arrest is often secondary to respiratory failure or shock; prioritize oxygenation & ventilation.
- Use appropriate compression depth & avoid leaning on chest between compressions.
- Ensure clear communication and delegation of tasks during resuscitation.
- If ALS back-up is delayed and hospital is nearby, consider transport while maintaining effective CPR.
- Consider reversible causes (4 Hs & 4 Ts) — hypoxia, hypovolemia, hypothermia, hypo/hyperkalemia, toxins, tamponade, tension pneumothorax, thrombosis.

References

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